

Evaluation of an AI-Assisted Vertical-Axis Wind Turbine Design

AI can organize research, generate CAD models, and help troubleshoot simulations, but using it well is an engineering challenge of its own. We investigated where AI tools can actually improve an engineering project while maintaining quality. Here is what we learned while designing a vertical axis wind turbine.

Vertical axis wind turbines (VAWTs) are an underdeveloped renewable-energy technology that is particularly promising for taking advantage of wind power in urban environments because they can handle multidirectional and turbulent wind conditions [1]. Renewable and distributed energy sources such as VAWTs could reduce strain on the centralized U.S. energy system and reduce carbon dioxide emissions. This would increase the U.S. energy system's resilience against natural disasters, and help mitigate the global warming that is making those disasters worse. [2] As a preliminary step toward designing a VAWT for a specific urban environment, we investigated and modeled a vertical axis wind turbine for our project.

Engineers are increasingly looking to AI tools to automate and accelerate repetitive tasks. AI tools have been rapidly developing and can now assist with documentation, formatting, CAD models, and simulation. AI can also be valuable for research and information organization, which is critical in the early stages of engineering design work. An example is Andrej Karpathy's LLM wiki, a structured knowledge base where an LLM organizes and maintains cross-linked Markdown documents from raw sources [3]. This makes an LLM wiki distinct from a general chatbot: instead of searching from scratch for every request, it collects and organizes easily accessible information in the wiki to build a usable knowledge base.

We are a team of two Engineering Science students at Smith College, working remotely for eight weeks with the support of our professor, Mike Kinsinger, and liaisons at 99P Labs: Duane Detwiler, Rajeev Chhajer, and Ryan Lingo. Over the course of this project, we tested several AI tools: Zoo Design Studio's Zookeeper and Fusion's Autodesk Assistant for CAD modeling, OpenCode with OpenAI models for our LLM wiki, and SimScale's chatbot, Ray, for computational fluid dynamics (CFD). These tools were able to save us time on tedious tasks but were not reliable for more detailed or complex work.

Process

Building the Wiki

We began by learning about and building our own LLM wiki. The wiki is built on a written contract defining how the agent should behave: it must rely on project sources, cite claims, flag unsupported material, and avoid common LLM pitfalls such as hallucination and sycophancy. We built an evaluation loop with questions designed to intentionally target these pitfalls and build our trust in its responses. Throughout the project, we evaluated the wiki and refined the contract to keep it reliable and useful. Our goal was to maintain a source-grounded workspace

that could help us find and sort information, locate knowledge gaps, and answer technical questions.

After the wiki was established, we began researching VAWTs and feeding information into the system. We created workflows to convert research PDFs into Markdown files while preserving text structure, figures, and equations, then had the agent generate wiki pages with information extracted from those sources. We organized the wiki pages into concepts, methods, designs, and parameters, which allowed us to find the information we needed. The design and parameter pages were particularly useful during concept selection. We selected important VAWT parameters such as cut-in speed, efficiency, and rated speed for the LLM to report at the top of each wiki design and parameter page. This made it easier to sort and rank designs with our agent and Obsidian, a Markdown-based note-taking app.

Throughout this process, we ran into missing images, poor formatting, unwanted edits to source text, slow or inconsistent runs, broken links, and the occasional inaccurate response. We addressed these problems by tightening the agent's skills, splitting detailed procedures into separate schema documents, adding Python-based image extraction, and improving our prompts. The result was a tool that could quickly synthesize complex material.

Defining our Design Goal

Once our LLM wiki was established, we selected a specific location for our proposed VAWT to narrow our design space and ensure simulations were realistic. To capitalize on the unique benefits of VAWTs, we looked for urban or suburban locations with moderate wind speeds and demand for renewable energy. However, our primary requirement was readily accessible wind data. Ultimately, we chose Boston Logan International Airport because of its accessible, high-resolution wind data and average wind speed of about 5 m/s [4]. Logan also has high energy demand and an interest in renewable energy, since the Massachusetts Port Authority intends to reach net-zero emissions by 2028 [5].

Before moving forward with BOS, we needed to ensure a VAWT could be installed on the premises, ideally near the center of the airfield where wind data is collected about 9 m above the ground. We used our LLM wiki agent to review Federal Aviation Administration regulations on airport construction [6]. It identified Governor's Island, an outer section of the airfield facing the bay on two sides, with at least 10 acres of VAWT-compatible space.

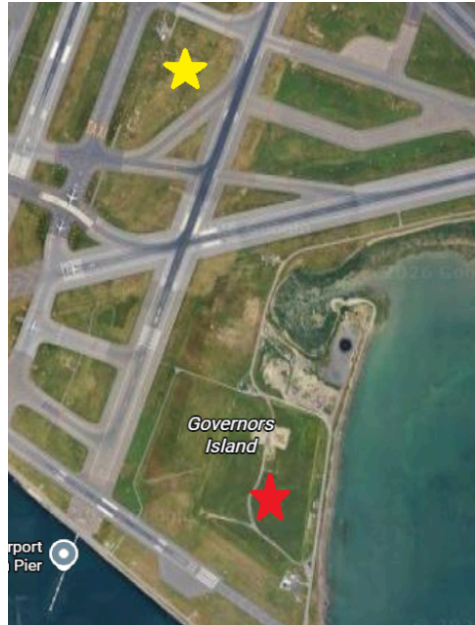


Figure 1: Wind data is collected at the location marked by the yellow star. Our proposed VAWT location on Governor's Island is marked by the red star.

With a location selected, our next step was to analyze the wind data and determine how the design could perform well in Governor's Island's wind conditions. We added five years of wind data to the wiki and prompted the agent to generate a histogram by wind speed.

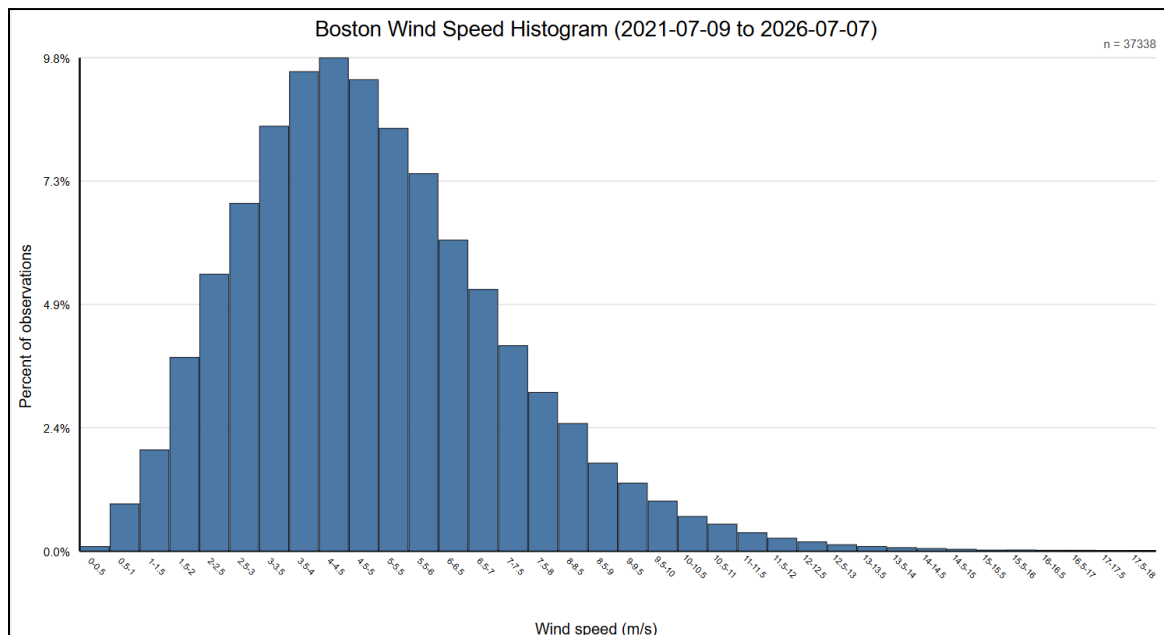


Figure 2: Agent-generated histogram of BOS wind data collected every hour from July 2021 - July 2026 [7].

The LLM quickly analyzed the data and, using its VAWT performance information, helped us identify design parameters like cut-in speed that could take advantage of available wind power at BOS.

Concept Selection

During concept selection, we determined which VAWT designs in the wiki were best suited to be starting points for optimization. To narrow our list of 63 VAWT designs, we each selected 8-10 promising designs based on their potential at BOS and our interest in pursuing them, with the help of the LLM wiki agent. After comparing our initial selections, we agreed on six designs to evaluate further. We developed selection criteria to score these designs: "Startup ability," "Efficiency," "Rated speed," "Cut-out speed," "Room for Growth" (the potential to iterate on the design), "CAD & Manufacturing Feasibility," and "Team Interest." First, we each manually scored the designs before asking the LLM wiki agent to repeat the exercise. The human and agent scores were very similar, especially for quantitative criteria. We replaced the agent scores for qualitative criteria, such as difficulty to model in CAD software or interest in pursuing a design, with the average of our scores. After scoring, we weighted the criteria based on their importance to our design goal and project feasibility, and ranked the designs.

To test score and weight sensitivity, we prompted the agent to rank the designs using three different weighting schemes based on 'personas': a beginner/student engineer with minimal CAD or CFD experience, an expert engineer with extensive CAD and CFD experience, and an engineer whose primary goal was to produce the best VAWT for BOS, regardless of constraints. The same two designs repeatedly came out on top, validating our selection process. This sensitivity test also clarified the relative strengths and weaknesses of the top two designs. Before moving on, we used the agent to apply the ranking system to all 63 designs in the wiki to ensure our initial pass hadn't prematurely eliminated any designs. H-type VAWT designs made up a large proportion of the top-ranked designs, and the previous best designs remained in the top two. The two designs selected to move forward were a dual-phase H-type VAWT [8] and a unique Darrieus VAWT [9], pictured below. To decide which design to begin testing, we moved both into the CAD phase to gauge modeling feasibility.

CAD Modeling

We explored two AI CAD tools: Zoo Design Studio's Zookeeper and Fusion's Autodesk Assistant. These tools let you generate parametric CAD models with natural text prompts. We used the LLM wiki to generate prompts for the CAD AI agents based on the two selected VAWT designs. The dual-phase H-type design was well described, so its prompt was robust and successful. The unique Darrieus design was more challenging because its source paper had incomplete geometry specifications. The source provided the overall rotor dimensions and described the inward-pointing, cup-like blade ends, but not blade profile or many structural dimensions. In contrast, the dual-phase H-type design provided complete geometric specifications for the turbine blades. That detail made it the stronger option for modeling, so we moved it into CFD analysis.

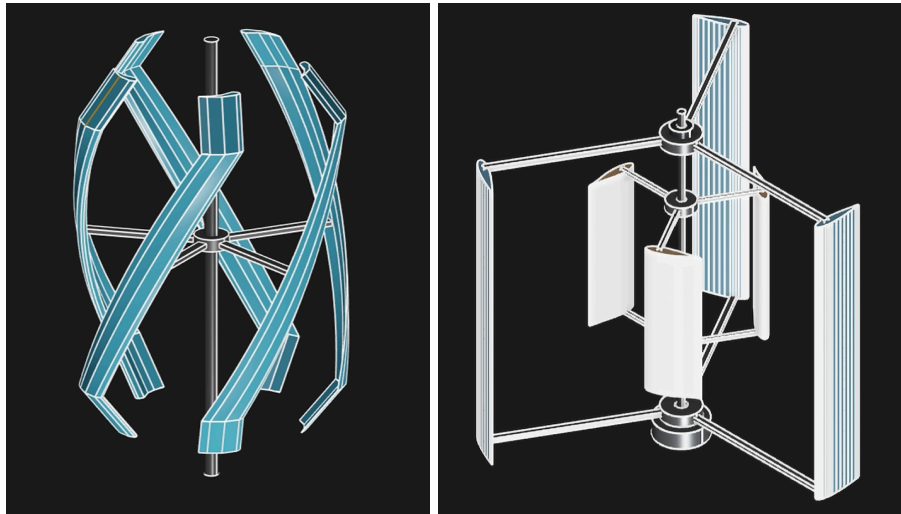


Figure 3: Zookeeper-assisted models of the unique Darrieus VAWT design (left) and dual rotor H-type VAWT design (right), completed in under 2 hours and 1 hour, respectively.

CFD Analysis

At the start of this project, our team had no experience with computational fluid dynamics (CFD). Our first step was to add resources to our LLM wiki so the agent could help teach us the basics of CFD and troubleshoot in SimScale, a multiphysics engineering simulation platform. After learning core CFD concepts, we began by attempting to replicate published airfoil simulation data using one of the airfoil profiles from our selected VAWT design. This required a significant trial and error, particularly with the mesh, and the LLM wiki agent's utility quickly plateaued as it began repeating advice we had already implemented or that was not relevant. After receiving advice from Dr. Phillip Aquino at HRI, we got much closer to our target. Because of time constraints, we moved on to simulating the full VAWT before successfully validating our airfoil simulation.

Preparing the mesh for the full VAWT simulation was much more complicated than the airfoil case and brought new challenges. With advice from Dr. Aquino, frequent 'chats' with Ray, SimScale's chatbot, and Google's Gemini, we learned more about proper mesh outcomes and settings and refinements to improve a mesh. During this process, we learned about structured and unstructured meshes (particularly SimScale's standard vs. hex-dominant meshing algorithms), key boundary-layer metrics like growth rate and relative thickness, the importance of tailored surface and volume refinements, and which geometry is easiest to mesh and which areas need simplification or heavy refinement. This resulted in a nearly acceptable mesh but no reliable torque or power results as the work was similarly cut short due to project time constraints.

AI Discussion

The AI tools we explored sped up tedious parts of this project and deepened our learning in some areas. However, none of these tools were completely reliable for more complex tasks so we had to consistently review their output. The LLM wiki was invaluable during our initial research and concept selection phases. It essentially functioned as a pseudo 'expert' on the sources we provided, letting us quickly retrieve and compare information across more than 60 sources with more confidence than a generic chatbot could provide. That reliability took significant work: defining the agent's behavior, evaluating its output, and adjusting its contract and skills. Even after we considered the wiki complete, we still needed to continually check its output and fine-tune it. That said, we found it also allowed us to shortcut the slower work of understanding new material, unless we intentionally took time to learn with the agent. It was easy to use the LLM wiki agent as an answering machine, especially since it behaved reliably. That reliability also made us more likely to trust its output completely, which highlighted the need for continual evaluation.

The LLM wiki significantly accelerated concept selection. The agent's information retrieval was extremely helpful for human-driven design exploration and AI-driven filters and rankings. The agent helped identify candidate concepts, extract structured data, compile comparison tables, and calculate weighted scores. This let us narrow a large set of designs to two in a single afternoon. However, the efficiency came at the cost of deeper learning through direct engagement with the material. So we used the wiki's quick retrieval while still manually assessing top designs, guiding the process with our own judgment, and reviewing every recommendation. This was especially important when sources were unclear or reported outcomes mainly through figures that the LLM struggled to ingest.

When using the LLM wiki for CFD assistance, we did not find the same reliability issues. Instead, the LLM wiki simply could not provide new, useful feedback. Initially, after ingesting SimScale's documentation and research papers about airfoil and VAWT CFD studies, the LLM wiki was helpful for getting our CFD work started. We used the agent to explain and quiz us on CFD concepts, settings, and guidelines, and asked for recommendations about mesh and simulation settings. However, as we continued to troubleshoot our airfoil simulations, the agent's advice became increasingly repetitive and random. We attempted to address this by adding a clause to the agent contract that allowed the agent to search the internet if it could not find an answer within the wiki, but this did not help as much as we had hoped. We instead turned to Ray, SimScale's chatbot, for specific questions about SimScale settings and VAWT-applicable recommendations. Ultimately, Ray and Google's Gemini provided a lot of assistance with setting up the full-turbine CFD simulation, but where they fell short, Dr. Aquino helped bridge the gap.

Our modeling process was greatly sped up by CAD AI agents, especially considering our limited previous experience with CAD. However, the process still required human review to check model validity and fine-tune details. We generated design prompts for the CAD agents using our LLM wiki agent and found that shorter, more constrained prompts worked better than the long-winded prompts that the LLM wiki agent initially produced. When requesting more

detailed edits such as adding strut-blade brackets or refining trailing edge shape, Zookeeper created disconnected blades, floating parts, and/or missing struts, while Autodesk Assistant often failed to execute changes and instead provided instructions for the user to manually complete them. Regardless of these late-stage modeling frustrations, the two tools saved us a lot of time by producing mostly correct, fairly complex models from a single prompt.

To make the most out of using AI in an engineering context, it is important to understand how it works and to learn to leverage its strengths and weaknesses. Doing so can elevate AI from a basic chatbot to an effective engineering tool that can be used across a range of applications. Different tasks require tailored prompts, contexts, and levels of human involvement in order to fully utilize its potential. Effective AI use is a skill that must be developed alongside engineering knowledge – not in replacement of it.

Next Steps

Given more time, we would have continued refining the VAWT through prototyping and testing of blade geometry, structural stability, and overall energy output. We also would have tested different turbine geometries in CFD to compare design changes more reliably. We were also interested in fully exploring SimScale's agentic tools. Under our SimScale plan, we had access to the basic chatbot, Ray. Although Ray was integrated directly into the platform, it could not read or edit simulation settings, which limited its usefulness. SimScale also offers two additional agentic tools: Physics AI, which can predict simulation results before they are run, and Engineering AI, which can automatically set up meshes and boundary conditions. These tools could have significantly accelerated our design and simulation workflow and are worth considering in a future project.

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